

Phylogenetic Analysis Using NCBI and Galaxy

GROUP 4 - Anatidae
BIO 300 - Cell & Molecular Biology

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Comparison with Published Studies

The phylogenetic tree generated from the COI sequences is generally consistent with recent studies on the evolutionary relationships within the family Anatidae. In the present study, *Anas platyrhynchos* and *Anas rubripes* formed the closest clade, indicating a high degree of genetic similarity. This agrees with the findings of Brown et al. (2022), who reported that closely related duck species share a recent common ancestry and exhibit strong genetic similarity despite occasional hybridization.

The tree also grouped *Mergus merganser* and *Lophodytes cucullatus* into the same clade, separate from the *Anas* species, reflecting their close evolutionary relationship within the merganser lineage. Similarly, Deef (2021) found that mitochondrial COI sequences effectively distinguished closely related duck species and produced phylogenetic groupings that corresponded with their evolutionary relationships.

Furthermore, the clear separation of *Phasianus colchicus* from all Anatidae species confirms its suitability as an outgroup for rooting the phylogenetic tree and determining the direction of evolutionary divergence. Overall, the topology obtained in this study agrees with previous molecular studies, supporting the use of the mitochondrial COI gene as a reliable marker for species identification and for reconstructing phylogenetic relationships among ducks.

Although the phylogenetic relationships observed in this study are consistent with previous molecular studies, it is important to recognize that using the COI gene alone has some limitations. As a single mitochondrial marker, COI reflects only maternal inheritance and may not fully represent the complete evolutionary history of species, particularly in groups where hybridization is common, such as ducks (Toews & Brelsford, 2012). Nevertheless, COI remains a reliable marker for species identification and phylogenetic analysis, and combining it with additional mitochondrial or nuclear genes could further improve the accuracy and resolution of evolutionary relationships (Hebert et al., 2003).